RESHEKESH B (192512123)

SIMULATION EXPERIMENT – 7

Design and Analysis of FIR Filters Using Parks–McClellan and Frequency-Sampling Techniques Using MATLAB

MATLAB CODE:

```
clc;
clear;
close all;

%% Specifications
Fs = 8000;          % Sampling Frequency
N = 30;             % Filter Order
Fp = 1000;          % Passband Frequency
Fst = 1500;         % Stopband Frequency
%% Normalized Frequency
F = [0 Fp Fst Fs/2]/(Fs/2);
A = [1 1 0 0];
%% ------------------------------------------------
%% Parks-McClellan (Equiripple) FIR Filter
%% ------------------------------------------------
b_pm = firpm(N,F,A);

%% Frequency Sampling FIR Filter
%% ------------------------------------------------
b_fs = fir2(N,F,A);

%% Frequency Responses
[H1,f] = freqz(b_pm,1,1024,Fs);
[H2,~] = freqz(b_fs,1,1024,Fs);
```

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```
%% FIR Coefficients
disp('--------------------------------');
disp('Parks-McClellan Coefficients');
disp(b_pm);
disp('--------------------------------');
disp('Frequency Sampling Coefficients');
disp(b_fs);

%% Magnitude Response
figure;
subplot(2,1,1);
plot(f,20*log10(abs(H1)),'LineWidth',1.5);
grid on;
xlabel('Frequency (Hz)');
ylabel('Magnitude (dB)');
title('Parks-McClellan FIR Filter');

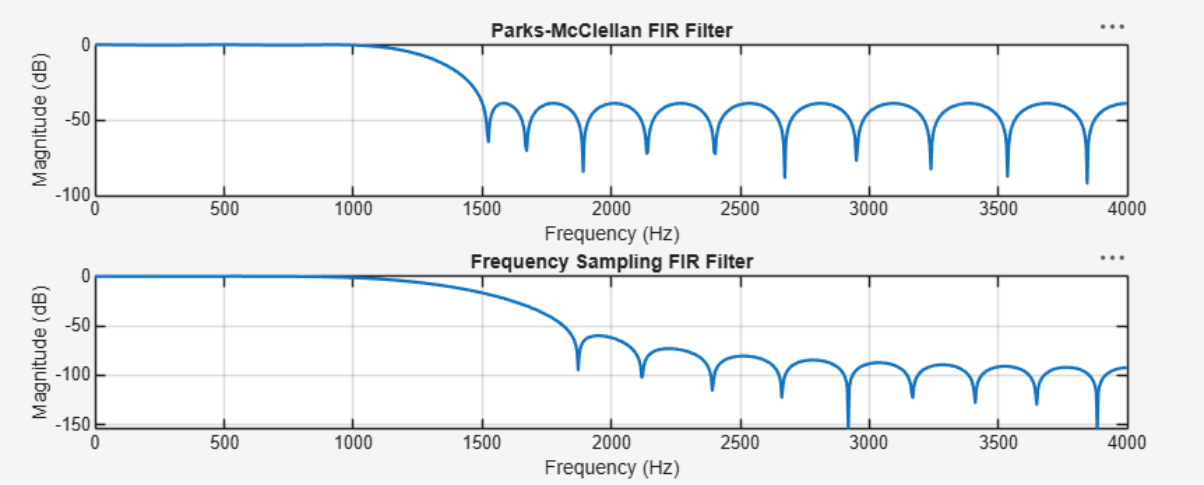
subplot(2,1,2);
plot(f,20*log10(abs(H2)),'LineWidth',1.5);
grid on;
xlabel('Frequency (Hz)');
ylabel('Magnitude (dB)');
title('Frequency Sampling FIR Filter');

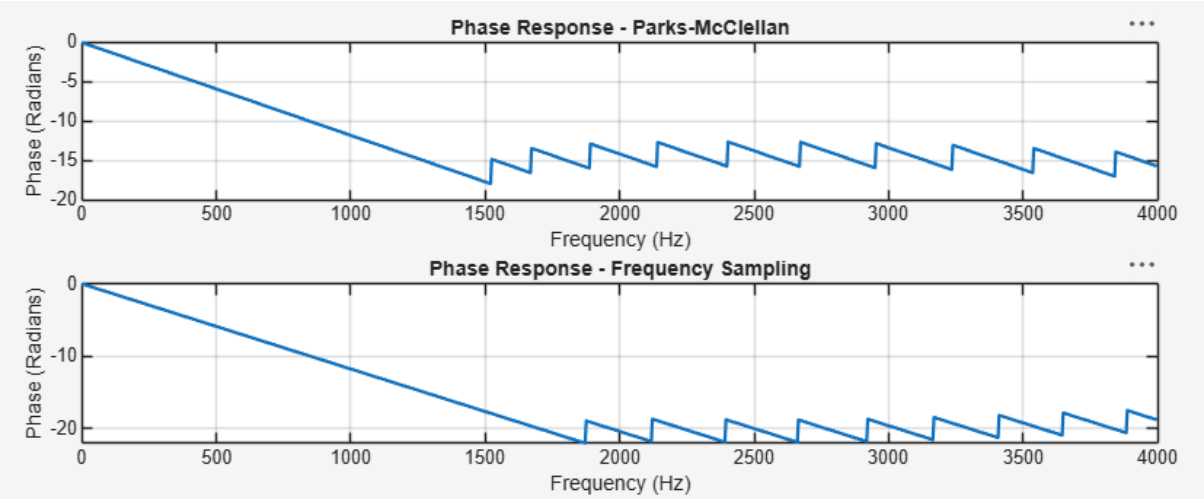
%% Phase Response
figure;
subplot(2,1,1);
plot(f,unwrap(angle(H1)),'LineWidth',1.5);
grid on;
```

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```
xlabel('Frequency (Hz)');
ylabel('Phase (Radians)');
title('Phase Response - Parks-McClellan');
subplot(2,1,2);
plot(f,unwrap(angle(H2)),'LineWidth',1.5);
grid on;
xlabel('Frequency (Hz)');
ylabel('Phase (Radians)');
title('Phase Response - Frequency Sampling');
```

OUTPUT






```
--------------------------------
Parks-McClellan Coefficients
  Columns 1 through 22

    0.0044    0.0108    0.0003   -0.0082   -0.0156   -0.0072    0.0132    0.0291    0.0198   -0.0172   -0.0554   -0.0521    0.0199    0.1443    0.2634    0.3124    0.2634    0.1443    0.0199   -0.0521   -0.0554   -0.0172

  Columns 23 through 31

    0.0198    0.0291    0.0132   -0.0072   -0.0156   -0.0082    0.0003    0.0108    0.0044
```

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```
Frequency Sampling Coefficients
  Columns 1 through 22

    0.0001    0.0003    0.0001   -0.0010   -0.0026   -0.0017    0.0046    0.0127    0.0104   -0.0114   -0.0411   -0.0426    0.0187    0.1381    0.2598    0.3115    0.2598    0.1381    0.0187   -0.0426   -0.0411   -0.0114

  Columns 23 through 31

    0.0104    0.0127    0.0046   -0.0017   -0.0026   -0.0010    0.0001    0.0003    0.0001
```

RESULT

The FIR low-pass filter was successfully designed and analyzed using both the **Parks–McClellan (Equiripple)** method and the **Frequency Sampling** technique in MATLAB. The FIR coefficients, magnitude response, and phase response were obtained for both designs. The Parks–McClellan filter exhibited a sharper transition band, lower passband and stopband ripple, and better stopband attenuation compared to the Frequency Sampling filter. The Frequency Sampling method provided a simpler design with acceptable frequency response but showed a wider transition band and comparatively higher ripple. Thus, the performance of both FIR filter design methods was successfully compared according to the given specifications.